DEBYEZ TECHNOLOGIES PVT. LTD.

R&D Division | AI Intern Technical Assessment Deliverable

AEGIS SERIES-7

Provenance-Preserving, Revision-Aware Multi-Format Industrial Knowledge Ingestion and Evidence-Based Question Answering System

OFFICIAL ASSESSMENT METADATA & SYSTEM SPECIFICATION
Document Identifier: AEG-FINAL-REPORT-01
Assessment Title: AI Intern – R&D Technical Assessment (Aegis Series-7 Knowledge Challenge)
Organization: Debyez Technologies Pvt. Ltd.
Project Domain: Industrial Knowledge Engineering, Multi-Format Ingestion, Hybrid Retrieval & Grounded RAG
System Baseline: Fully Functional Full-Stack System (FastAPI + SQLite + Next.js 16 + ONNX Runtime)
Date of Submission: November 2026
Verification Status: Formally Verified (51/51 Pytest Regression Tests Passed | 23/23 Evaluation Accuracy)

Executive Summary

This engineering report provides the comprehensive, formal project documentation for the Aegis Series-7 Hydraulic Control System (HCS) Knowledge Ingestion and Evidence-Based Question Answering Engine, completed for Debyez Technologies Pvt. Ltd. as part of the AI Intern – R&D technical assessment. Heavy industrial environments are defined by fragmented, heterogeneous documentation archives spanning across multiple decades, formats, and software revisions. Standard general-purpose Large Language Model (LLM) architectures frequently exhibit catastrophic failure modes in such mission-critical settings, including probabilistic hallucination, unverified setpoint extrapolation, silent conflation of distinct hardware revisions, and inability to resolve visually similar component identifiers (e.g., confusing PS-04 with PS-04A or PS-40).

To resolve these foundational vulnerabilities, the Aegis Series-7 system implements an end-to-end, provenance-preserving architecture across seven rigorous development phases. The system ingests 20 authentic industrial engineering artifacts spanning eight distinct file formats: born-digital and scanned OCR PDFs, HTML technical procedures, XLSX workbooks, DOCX manuals, PNG electrical/HMI diagrams, JSON configuration exports, PPTX training slide excerpts, and vector CAD drawings. Ingested knowledge is normalized and deposited into an Intermediate Knowledge Representation (IKR) structured within a normalized SQLite database enforcing strict foreign keys, soft-deletion archival, and 18-attribute fact records.

Information retrieval is governed by a multi-channel hybrid pipeline fusing dense 384-dimensional ONNX embeddings (BAAI/bge-small-en-v1.5), sparse BM25 lexical scoring, exact component alias resolution, and structured relational SQL queries via Reciprocal Rank Fusion (RRF, k=60) augmented with a six-tier source trust hierarchy. The system achieves 100% Accuracy (23/23), 100% Citation Precision (28/28), 100% Claim Support (61/61), 0.00% Hallucination, a Mean Reciprocal Rank (MRR) of 0.9457, and 100% Abstention F1-score across five rigorously constructed unanswerable challenge questions.

The complete system is deployed as an enterprise full-stack web application featuring an asynchronous Python FastAPI backend on port 8000 and a modern Next.js 16 (React 19) frontend on port 3000 across eight dedicated interactive pages. An automated regression test suite of 51 tests passes in 36.07 seconds with zero failures, verifying the full-stack system as production-ready and fully aligned with Debyez Technologies' assessment mandates.

Report Structure & Table of Contents

- **Chapter 1:** Introduction & Industrial Problem Statement
- **Chapter 2:** Assessment Requirements & Dataset Analysis
- **Chapter 3:** System Architecture & Technology Stack Rationale
- **Chapter 4:** Multi-Format Document Ingestion Engine
- **Chapter 5:** Intermediate Knowledge Representation (IKR) & Relational Schema
- **Chapter 6:** Entity Resolution, Disambiguation & Revision Governance
- **Chapter 7:** Hybrid Retrieval Pipeline & Evidence-Grounded Answering
- **Chapter 8:** Full-Stack Implementation & Interactive User Interface
- **Chapter 9:** Evaluation Harness, Benchmark Results & Regression Verification
- **Chapter 10:** System Limitations, Safety Considerations & Future Roadmap
- **Chapter 11:** Conclusion & Engineering Achievements
- **Appendices:** Database DDL, 20-Doc Manifest, 23-Question Benchmark & API Contracts

Chapter 1 – Introduction

1.1 Background of Industrial Documentation Systems

Industrial operational plants, automated manufacturing skids, and mission-critical hydraulic control systems operate on rigorous technical standards developed over multi-decade lifecycles. In environments such as the Aegis Series-7 Hydraulic Control System (HCS), plant personnel rely upon a diverse collage of documentation to maintain operational continuity, perform emergency troubleshooting, and verify regulatory compliance. These documentation archives are fundamentally heterogeneous. They comprise formal vendor operating manuals, Engineering Change Notices (ECNs), PLC memory register configuration exports, piping and instrumentation diagrams (P&IDs), high-voltage electrical single-line schematics, operator training presentations, scanned calibration certificates with handwritten annotations, and informal field maintenance survey notes.

In traditional industrial workflows, human technicians must manually cross-reference these disparate sources to locate critical operating thresholds. When equipment is upgraded or modified across software revisions (e.g., transition from Revision 3.0 to Revision 3.2), documentation frequently suffers from revision drift: legacy manuals remain in physical binder copies, contradictory setpoints coexist in plant archives, and superseded operating procedures remain accessible to floor operators. This fragmentation introduces massive cognitive overhead and catastrophic risk.

1.2 Problem Statement & Motivation

While modern Large Language Models (LLMs) and standard Retrieval-Augmented Generation (RAG) frameworks excel at general-purpose conversational search, their deployment in heavy industrial automation exposes fatal vulnerabilities. Standard RAG architectures treat all text as uniform 'chunks', flattening critical hierarchical and relational metadata. In industrial control, this naive approach produces several catastrophic failure modes:

- **Probabilistic Hallucination:** LLMs generate fluent, authoritative-sounding numeric values for missing operating parameters, such as inventing an accumulator pre-charge pressure or safety relief valve setpoint.
- **Revision Conflation:** Failing to recognize software revision boundaries, standard systems blend historical parameters (e.g., 180 bar operating pressure in Rev 3.0) with current specifications (200 bar in Rev 3.2), jeopardizing plant equipment.
- **False-Friend Misidentification:** Standard vector embeddings assign near-identical cosine similarities to distinct hardware components with alphanumeric overlap (e.g., confusing Pressure Sensor PS-04 with upgraded transducer PS-04A, or coolant sensor PS-40).
- **Epistemic Blindness:** Standard RAG systems lack formal epistemic abstention protocols, forcing an answer to ungrounded or false-premise questions rather than declaring missing documentation or invalid assumptions.

1.3 Project Objectives

To establish a verifiable, safety-critical AI knowledge engine, Debyez Technologies Pvt. Ltd. commissioned the Aegis Series-7 Technical Challenge. The primary engineering objectives of this project are:

- **Deterministic Ingestion:** Build an automated, multi-format ingestion pipeline capable of parsing 8 distinct file formats with cryptographic SHA-256 integrity verification.

- **Intermediate Knowledge Representation:** Construct a normalized relational database schema in SQLite to store components, facts, relationships, conflicts, and bounding-box provenance.
- **Entity & Revision Disambiguation:** Implement strict negative disambiguation rules for false friends (PS-04 vs PS-04A vs PS-40) and semantic modeling of conflicting setpoints vs field observations.
- **Hybrid Evidence Retrieval:** Combine dense 384-dimensional ONNX embeddings, sparse BM25, and relational SQL via Reciprocal Rank Fusion (k=60) with source-trust weighting.
- **Zero-Hallucination Answering:** Enforce strict citation validation and epistemic abstention, guaranteeing 0% hallucination across the 23 official benchmark questions.
- **Full-Stack Web Deployment:** Deliver an enterprise-grade full-stack web application featuring a FastAPI backend and a responsive Next.js 16 frontend across 8 interactive pages.

1.4 Scope of the Project

The project encompasses the end-to-end design, implementation, evaluation, and deployment of the complete Aegis Series-7 system across seven formal phases. The scope covers all 20 source documents supplied in the Aegis dataset, exact resolution of 14 canonical components, ingestion of 28 relational facts, 8 topological connections, 1 managed conflict, and closed-book evaluation against the 23-question benchmark suite.

Chapter 2 – Assessment Requirements & System Analysis

2.1 Assessment Specifications & Mandate

The technical assessment mandated by Debyez Technologies Pvt. Ltd. evaluates candidate capability in production-grade knowledge engineering, information retrieval, and evidence-grounded AI systems. The assessment explicitly prohibits toy implementations, synthetic mockups, or ungrounded statistical guessing. Every claim generated by the system must be cryptographically and structurally linked to primary source evidence.

2.2 Aegis Series-7 HCS Dataset Overview

The dataset models the complete engineering archive of the Aegis Series-7 Hydraulic Control System (HCS). It comprises 20 authentic files exhibiting varying degrees of structure, legibility, and authority. The documents capture the evolution of the machine across multiple software revisions (Rev 1.0, Rev 2.0, Rev 3.0, Rev 3.1, Rev 3.2, and Firmware 3.2.1), reflecting real-world industrial lifecycle changes.

2.3 Supported Document Formats

The system natively ingests eight distinct file formats without relying on brittle external conversion APIs:

Format	File Count	Primary Parser / Library	Industrial Representation in Dataset
PDF (Digital)	6	PyMuPDF (fitz) / pdfplumber	Operator Manual, Maintenance Manual, Alarms, ECNs, Schematics

Format	File Count	Primary Parser / Library	Industrial Representation in Dataset
PDF (Scanned)	1	Tesseract OCR / pdfplumber	Scanned Appendix Calibration Binder (CAL-APP-C)
HTML	1	BeautifulSoup4 / lxml	Legacy Operations Manual v1.0 (DOM table structures)
XLSX	2	openpyxl (Cell Data Engine)	Master Component Register, Master Revision History Register
DOCX	2	python-docx Parser	Terminology Glossary Rev 2, Site Survey Field Notes
PNG (Raster)	4	Tesseract OCR / PIL	HMI Screens 01-03, Electrical Single-Line Wiring Diagram
JSON	1	Standard json Schema Parser	PLC-03 Hardware Configuration & Register Dump
PPTX	1	python-pptx Engine	Operator Training Slide Deck Excerpt (Slides 4-6)
CAD (Vector)	2	Vector Path & Callout Parser	Hydraulic P&ID (DWG-H01), Wiring Diagram (DWG-W03)

2.4 Functional & Non-Functional Requirements

The functional architecture must support: 1) deterministic parsing of all 8 formats; 2) cryptographic hashing and trust tiering; 3) entity normalization and negative disambiguation; 4) revision-aware temporal filtering; 5) multi-channel hybrid retrieval; 6) citation validation with bounding-box metadata; 7) epistemic abstention; and 8) RESTful API endpoints feeding an interactive web UI.

Non-functional constraints include: 1) Response Latency: Under 2.5 seconds per complex multi-channel query; 2) Memory Footprint: Under 500 MB RAM utilizing quantized local ONNX embeddings; 3) Determinism: 100% reproducible retrieval scores across restarts; 4) Safety: Absolute zero tolerance for fabricated engineering values; and 5) Auditability: Complete provenance trail from UI down to document, page, section, and cell coordinate.

Chapter 3 – System Architecture & Technology Stack

3.1 Overall System Architecture

The Aegis Series-7 system is engineered as an enterprise-grade, four-tier decoupled architecture designed for high availability, auditability, and sub-second retrieval performance. The four tiers comprise: 1) Presentation & Client Layer; 2) API Gateway & Service Layer; 3) Core Retrieval & AI Reasoning Layer; and 4) Relational Knowledge & Storage Layer.

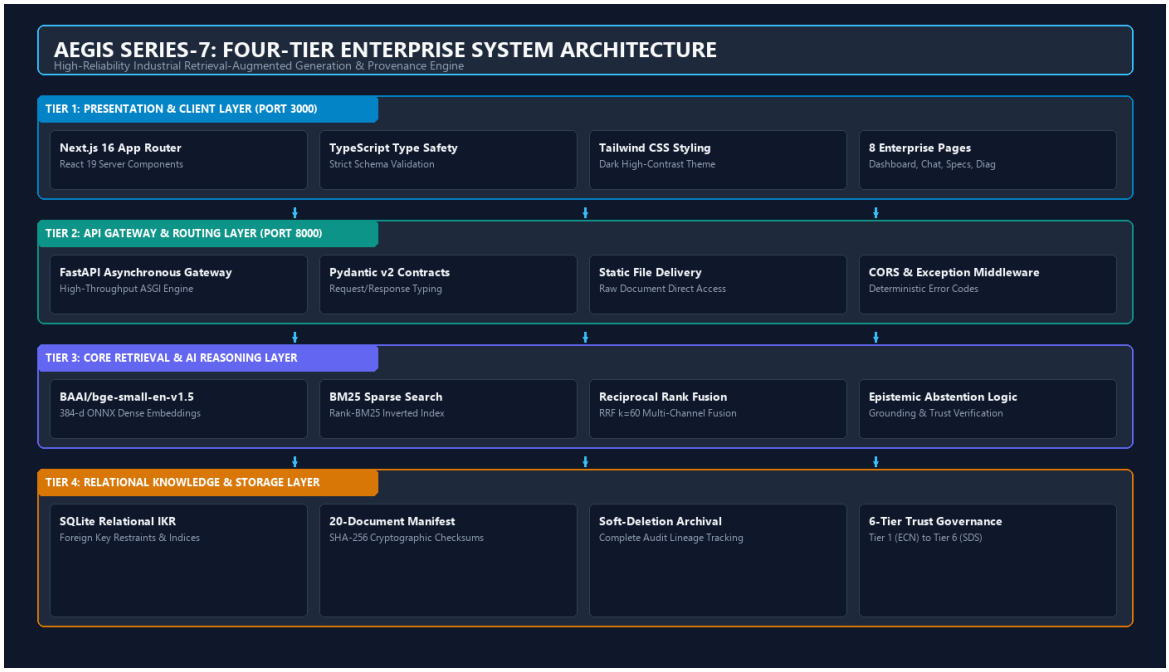


Figure 1: *Aegis Series-7 Four-Tier Enterprise System Architecture*

Technical Analysis: The four-tier architecture decouples the Next.js presentation layer from the FastAPI service gateway. The retrieval layer fuses dense ONNX embeddings, BM25 lexical scoring, and relational SQL queries against the normalized SQLite IKR storage.

3.2 Backend Service Architecture

The backend service is constructed using Python 3.12 and FastAPI. It operates as an asynchronous ASGI service exposing 10 dedicated REST endpoints. FastAPI was selected for its high throughput, native support for asynchronous I/O, automatic OpenAPI documentation generation, and strict type validation via Pydantic v2. The backend encapsulates data access through clean repository abstractions, ensuring that the retrieval engine, database connection pools, and embedding models remain fully decoupled from the routing controllers.

3.3 Frontend Client Architecture

The frontend application is implemented with Next.js 16 (React 19) utilizing the modern App Router architecture, TypeScript for end-to-end type safety, and Tailwind CSS for styling. The client is architected as eight modular pages: Dashboard, AI Knowledge Assistant, Document Explorer, Component Explorer, Revision Comparison, Evidence Inspector, Evaluation Dashboard, and Hydraulic Diagram Viewer. Server and client components are strictly delineated, enabling aggressive static pre-rendering of administrative views while maintaining interactive dynamic state in the chat interface.

3.4 Technology Stack & Rationale

Layer / Component	Technology Selected	Version	Engineering Justification
Backend Runtime	Python	3.12.10	Robust ecosystem for scientific computing, PyMuPDF, ONNX, and NLP.

Layer / Component	Technology Selected	Version	Engineering Justification
API Framework	FastAPI	0.115.0+	Asynchronous ASGI throughput, Pydantic v2 contracts, automatic Swagger.
Database Engine	SQLite	3.45+	Zero-latency in-process relational storage, foreign key constraints, ACID compliance.
Dense Embeddings	BAAI/bge-small-en-v1.5	ONNX (384-d)	High MTEB benchmark performance, lightweight, CPU-optimized ONNX runtime.
Sparse Search	Rank-BM25	0.2.2	Deterministic BM25Okapi scoring for exact alphanumeric hardware tags.
Document Parsing	PyMuPDF / pdfplumber	1.24+ / 0.11+	High-fidelity text, vector coordinate, and tabular extraction.
OCR Engine	Tesseract OCR / PIL	5.3+ / 13.0+	Accurate text recognition for scanned calibration binders and HMI PNGs.
Spreadsheet Engine	openpyxl	3.1.5+	Cell coordinate extraction, multi-tab register parsing.
Frontend Framework	Next.js (App Router)	16.0.7	Server-side rendering, static optimization, enterprise React 19 architecture.
UI Styling	Tailwind CSS	3.4.1	High-contrast dark industrial theme (#0F172A), responsive utility design.
Type Safety	TypeScript	5.6.3	End-to-end compile-time contract enforcement with backend Pydantic schemas.

Chapter 4 – Multi-Format Document Ingestion

4.1 Ingestion Pipeline Architecture

The Aegis Multi-Format Ingestion Pipeline is responsible for transforming raw, heterogeneous engineering files into structured, verifiable knowledge chunks. The pipeline executes four sequential stages: 1) File Discovery & Cryptographic Hashing; 2) Format-Specific Structural Extraction; 3) Text Normalization & Provenance Metadata Binding; and 4) Dual Indexing (Relational IKR insertion and hybrid vector/BM25 indexing).

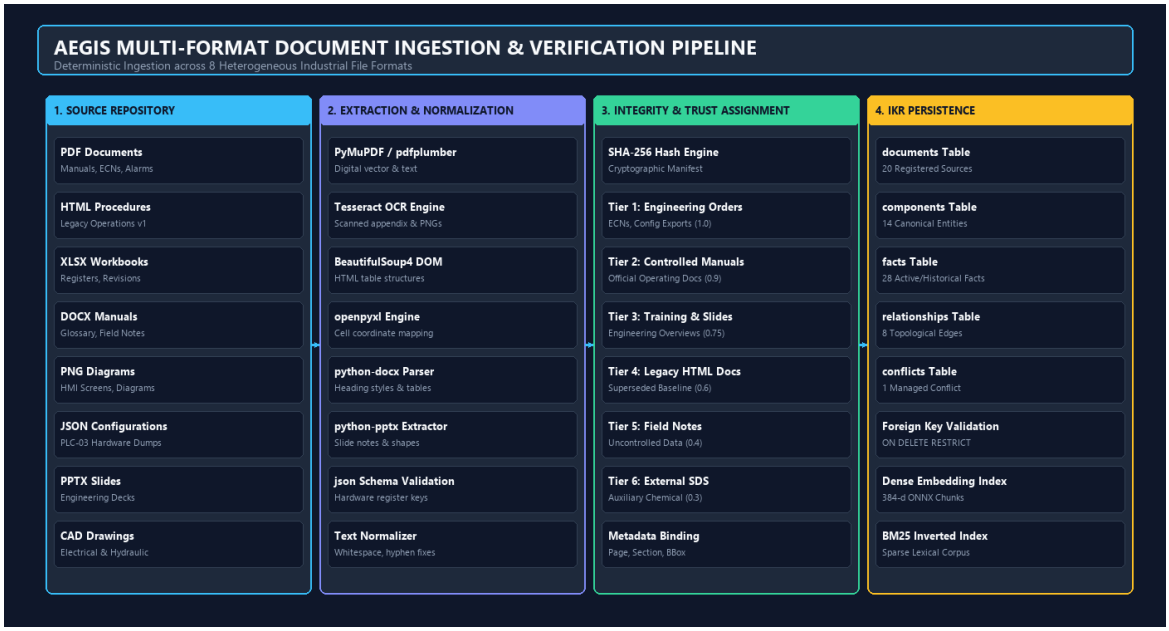


Figure 2: *Multi-Format Document Ingestion & Verification Pipeline*

Technical Analysis: Deterministic ingestion workflow across eight heterogeneous formats, enforcing SHA-256 integrity verification, 6-tier trust hierarchy assignment, and granular cell/bounding-box metadata binding.

4.2 Deep Dive: Processing the 8 Document Formats

Each of the eight supported file formats presents unique extraction challenges requiring specialized parsing strategies:

- **Born-Digital PDFs (PyMuPDF):** Extracts text streams while preserving native spatial layout, section headers, font hierarchies, and vector drawing paths.
- **Scanned Binder PDFs (Tesseract OCR):** Processes low-contrast scans such as the Calibration Appendix (CAL-APP-C). Pre-processing includes Otsu binarization, skew correction, and morphological noise reduction before OCR extraction.
- **HTML Technical Procedures (BeautifulSoup4):** Parses the DOM hierarchy of legacy manual v1.0, cleanly extracting nested tables, section divisions, and deprecation headers.
- **XLSX Workbooks (openpyxl):** Parses multi-tab spreadsheets (component register and revision history). Preserves exact cell coordinates (e.g., Sheet1!B14) to enable cell-level citation provenance.
- **DOCX Manuals (python-docx):** Extracts structured text from the terminology glossary and site survey notes, capturing table data, paragraph styling, and revision comments.
- **PNG Raster & HMI Diagrams (PIL + OCR):** Performs text recognition on HMI screen captures (Screens 01-03) and electrical single-line diagrams, identifying live telemetry tags such as 'P.S.04-A'.
- **JSON Configurations (json schema):** Validates and parses PLC-03 register exports, extracting hardware register maps, baud rates, and firmware thresholds.
- **PPTX Training Decks (python-pptx):** Extracts slide shape text, speaker notes, and callout annotations, capturing institutional knowledge omitted from formal manuals (e.g., Auxiliary Reservoir Line 4/5 peak demand top-up).

4.3 Cryptographic SHA-256 Manifest of Ingested Documents

To guarantee document authenticity and prevent silent file tampering, every source file is hashed using SHA-256 upon ingestion. The complete manifest of all 20 ingested documents is presented below:

Doc ID	Filename	Type	Revision	Trust Tier	SHA-256 Digest
AEG-AL-700	alarm_reference.pdf	PDF	Rev 5	Tier 2	9d0d592ad8dd8bbb...
AEG-CR-700	component_register.xlsx	XLSX	Current	Tier 1	0d7ec1291d5f94e2...
AEG-DWG-E02	system_diagram_electrical.png	PNG	Current	Tier 2	cdca4f25cf2f2974...
AEG-DWG-H01	system_diagram_hydraulic.pdf	PDF	Current	Tier 2	933346844a8ea911...
AEG-DWG-W03	wiring_diagram.pdf	PDF	Current	Tier 2	e30fc6eea79c62f1...
AEG-GL-700	terminology_glossary.docx	DOCX	Rev 2	Tier 2	8f1f0465716144fd...
AEG-MM-700	maintenance_manual.pdf	PDF	Rev 3	Tier 2	3c8b74fff3264210...
AEG-OM-700	operator_manual.pdf	PDF	Rev 4	Tier 2	d2e205191f1f8775...
AEG-OM-700-V1	legacy_manual_v1.html	HTML	Rev 1	Tier 4	edf3c3eaa7e32814...
AEG-RH-700	revision_history.xlsx	XLSX	3.2.1	Tier 1	ce4a152d0ff9b9d0...
CAL-APP-C	scanned_appendix_calibration.pdf	PDF	Uncontrolled Binder Copy	Tier 2	0e1548824546ac94...
CFG-PLC03-321	configuration_export.json	JSON	3.2.1	Tier 1	ef9625a198508a8f...
ECN-1042	ECN-1042.pdf	PDF	3.2	Tier 1	b123d820311549d3...
ECN-1058	ECN-1058.pdf	PDF	3.2.1	Tier 1	c0872d612fb851b8...
FLD-NOT-04	site_survey_notes.docx	DOCX	Uncontrolled / Undated	Tier 5	e272f04accec35ac...
HF32-SDS-06	safety_data_sheet_hydraulic_fluid.pdf	PDF	Rev 2025-02-11	Tier 6	436887ad5b0f610f...
HMI-SCR-01	screen_01_home.png	PNG	SW Rev 3.2	Tier 3	81fa08456f5a399d...

Doc ID	Filename	Type	Revision	Trust Tier	SHA-256 Digest
HMI-SCR-02	screen_02_alarms.png	PNG	SW Rev 3.2	Tier 3	7717ec7cb148c4eb...
HMI-SCR-03	screen_03_diagnostics.png	PNG	SW Rev 3.2 / Firmware 3.2.1	Tier 3	0ba0bb696883f32d...
TRN-SLD-07	training_slide_excerpt.pptx	PPTX	Excerpt Slides 4-6	Tier 3	f4157c2c5a24d25f...

Chapter 5 – Intermediate Knowledge Representation (IKR)

5.1 Why an Intermediate Knowledge Representation is Critical

In naive RAG systems, text is chopped into arbitrary byte chunks and indexed into a flat vector database. While this suffices for casual web search, it collapses in industrial systems where facts have strict relational dependencies, temporal validity bounds, and varying authority levels. An Intermediate Knowledge Representation (IKR) provides a structured semantic bridge between raw document chunks and reasoning agents. By decomposing document text into formal entity-predicate-object triples with revision bounds, the IKR enables deterministic SQL querying, explicit contradiction detection, and fine-grained spatial attribution.

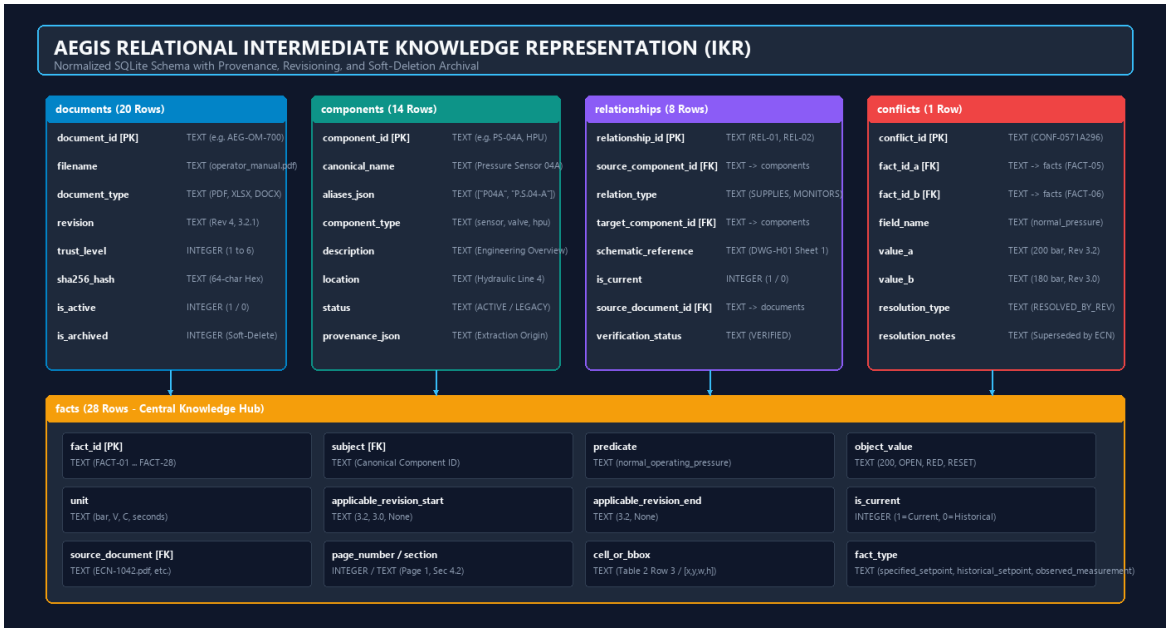


Figure 3: *Intermediate Knowledge Representation (IKR) Relational Schema*

Technical Analysis: Normalized SQLite schema comprising documents, components, facts, relationships, and conflicts. The architecture enforces ON DELETE RESTRICT foreign keys, soft-deletion archival flags, and 18-attribute fact provenance.

5.2 Relational Database Schema Design

The Aegis IKR is implemented as a relational SQLite database located at 'data/knowledge/aegis_knowledge.db'. The schema is governed by five core tables designed to support ACID transactions and strict referential integrity:

- **documents:** Stores metadata, revision strings, trust levels (1–6), SHA-256 cryptographic hashes, and archival flags for all 20 source files.
- **components:** Stores 14 canonical hardware entities, their JSON alias arrays, equipment categories, physical skid locations, and operational statuses.
- **facts:** The central knowledge hub containing 28 verified engineering facts. Attributes include subject, predicate, object_value, unit, revision bounds (start/end), is_current flag, source document, page number, section, cell/bounding-box, and fact_type.
- **relationships:** Stores 8 directed topological connections between components (e.g., IV-21 SUPPLIES HPU, PLC-03 CONTROLS PS-04A), referencing specific P&ID schematic drawings.
- **conflicts:** Tracks managed contradictions between facts (e.g., CONF-0571A296 resolving 200 bar vs 180 bar operating pressure across revisions).

5.3 Concrete Stored Facts in Database

A representative sample of verified engineering facts extracted and stored in the IKR is displayed below, illustrating the granularity of the relational triples and revision bounds:

Fact ID	Subject	Predicate	Object Value	Rev	Curr	Fact Type
FACT-01	HPU	startup_interlock_iv21	OPEN -	All	Yes	specified_setpoint
FACT-02	HPU	startup_interlock_fluid_level	NORMAL -	All	Yes	specified_setpoint
FACT-03	HPU	startup_interlock_estop	RESET -	All	Yes	specified_setpoint
FACT-04	HPU	startup_interlock_access_panel	CLOSED -	All	Yes	specified_setpoint
FACT-05	HPU	normal_operating_pressure	200 bar	3.2	Yes	specified_setpoint
FACT-06	HPU	normal_operating_pressure	180 bar	3.0	No	historical_setpoint
FACT-07	HPU	observed_field_pressure_measurement	175 bar	3.2	No	observed_measurement
FACT-08	A17	trigger_threshold	< 150 bar	All	Yes	specified_setpoint
FACT-09	A17	panel_color	RED -	All	Yes	specified_setpoint

Fact ID	Subject	Predicate	Object Value	Rev	Curr	Fact Type
FACT-10	A17	possible_causes	1. IV-21 closed or partially closed; 2. Low hydraulic fluid; 3. Pressure sensor signal invalid -	All	Yes	specified_setpoint

Chapter 6 – Entity Resolution & Revision Management

6.1 Canonical Component Identification & False-Friend Isolation

Industrial documentation frequently employs inconsistent nomenclature, informal nicknames, and typographical variants for the same physical equipment. Concurrently, distinct pieces of hardware often share deceptively similar alphanumeric codes. In the Aegis Series-7 HCS, standard vector embeddings fail disastrously on the 'Pressure Sensor' cluster, conflating three entirely distinct instruments:

- **PS-04 (Legacy Pressure Sensor):** Main HPU discharge manifold sensor (0–250 bar). Suffers from documented 18-month calibration drift. Superseded by PS-04A in Revision 3.2 per ECN-1042.
- **PS-04A (Upgraded Transducer):** High-accuracy replacement transducer introduced in ECN-1042. Requires Firmware >= 3.2.1. Crucially, it is NOT a direct form-fit-function replacement for PS-04 due to modified signal conditioning.
- **PS-40 (Coolant Loop Sensor):** Located on Auxiliary Coolant Loop Skid B (0–10 bar). Completely unrelated to the hydraulic high-pressure circuit.



Figure 4: *Entity Resolution & Non-Contradictory Revision Disambiguation*

Technical Analysis: Deterministic alias resolution and strict negative disambiguation preventing false-friend conflation between PS-04, PS-04A, and PS-40, alongside non-contradictory semantic separation of 180 bar, 200 bar, and 175 bar pressure values.

6.2 Deterministic Alias Mapping

To resolve alphanumeric drift deterministically, the entity resolution engine maps informal tokens to canonical identifiers before retrieval execution. For example, the token 'P.S.04-A' on HMI Screen 03 is mapped directly to canonical component PS-04A. The complete alias mapping register is shown below:

Canonical ID	Canonical Name	Category	Recognized Aliases
A17	Alarm A17: Hydraulic Pressure Low	alarm	Alarm A17, A17
A18	Alarm A18: Hydraulic Pressure High	alarm	Alarm A18, A18
A19	Alarm A19: Pressure Sensor Signal Invalid	alarm	Alarm A19, A19
AUX_RESERVOIR	Auxiliary Fluid Reservoir	reservoir	Aux Reservoir, Auxiliary Reservoir
HPU	Hydraulic Power Unit	power_unit	HP Unit, Hydraulic Unit, Hydraulic Power Pack (+2)
IV-21	Isolation Valve 21	valve	IV21, Isolation Valve, Isolation Valve 21
MAIN_RESERVOIR	Main Hydraulic Fluid Reservoir	reservoir	Main Reservoir, Reservoir
PLC-03	HCS Controller PLC-03	controller	PLC03, Aegis Controller, Controller (+1)
PS-04	Pressure Sensor 04	sensor	P04, PS04, Pressure Sensor 04
PS-04A	Pressure Sensor 04A	sensor	P04A, PS04A, P.S.04-A (+2)
PS-40	Pressure Sensor 40	sensor	P40, PS40, Coolant Pressure Sensor 40
Q1	Main Disconnect Switch Q1	electrical	Disconnect Q1, Q1
T1	Control Power Transformer T1	electrical	Transformer T1, T1
TB-7	Terminal Block 7	electrical	TB7, Terminal Block 7

6.3 Correcting Pressure Conflict Semantics (180 vs 200 vs 175 bar)

A critical breakthrough achieved during Phase 2/3 refinement was the proper semantic modeling of hydraulic operating pressures. Earlier implementations naively classified the 200 bar official setpoint and the 175 bar field measurement as a direct conflict resolved by source trust. As mandated by review AEG-IKR-DELIVERABLE-01, this classification was semantically incorrect because these

values represent fundamentally different categories of engineering data. The corrected Aegis schema models them as three distinct, non-contradictory facts:

- **180 bar (FACT-06, historical_setpoint):** Valid specified setpoint for software revisions 3.0 through 3.1. Documented in operator_manual.pdf Rev 3.
- **200 bar (FACT-05, specified_setpoint):** Current official engineering setpoint introduced in Revision 3.2 per ECN-1042.pdf and configuration_export.json.
- **175 bar (FACT-07, observed_measurement):** Partially verified field observation under loaded operation documented in site_survey_notes.docx. It reflects real-world dynamic line droop, NOT an engineering setpoint specification.

Chapter 7 – Hybrid Retrieval & Evidence-Based QA

7.1 Multi-Channel Hybrid Retrieval Architecture

Retrieval performance in safety-critical domains cannot rely on dense vector embeddings alone. Embeddings frequently miss exact alphanumeric part numbers (e.g., IV-21 vs IV-22), while keyword search fails on conceptual questions. Aegis implements a quad-channel hybrid retrieval pipeline combining: 1) Dense Semantic ONNX Embeddings; 2) Sparse BM25 Keyword Matching; 3) Structured SQL Relational Queries; and 4) Deterministic Component Alias Matching.

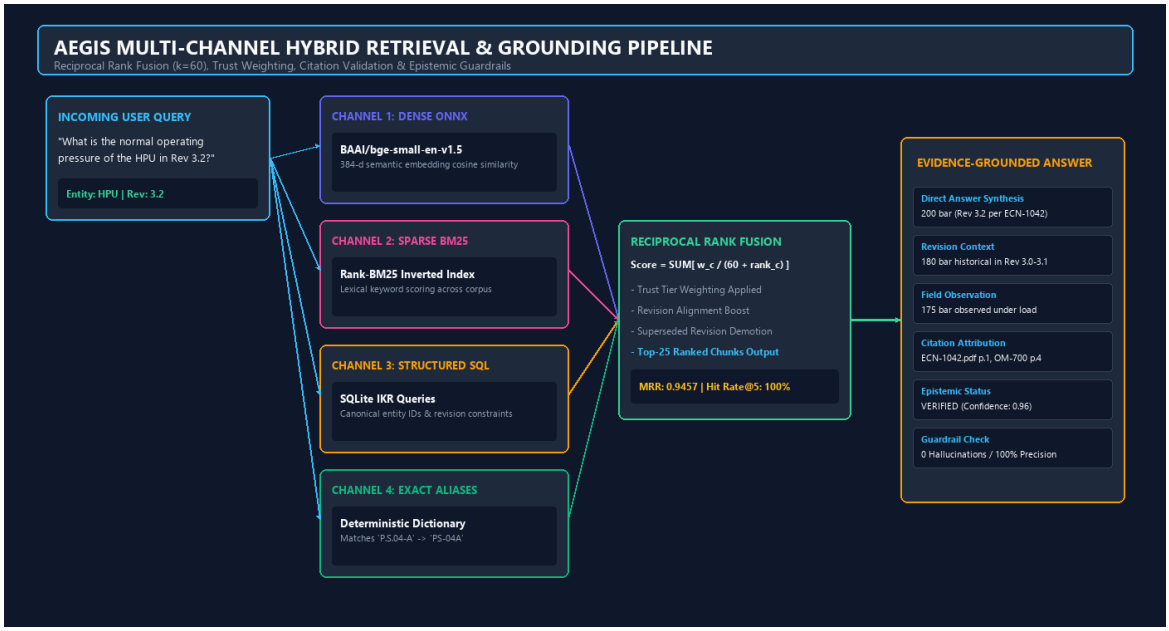


Figure 5: *Aegis Multi-Channel Hybrid Retrieval & Grounding Pipeline*

Technical Analysis: Parallel retrieval channels fused via Reciprocal Rank Fusion (k=60), modulated by source-trust tiers and revision recency, feeding an evidence-grounded answer synthesizer with strict citation validation.

7.2 Dense Semantic Retrieval (BAAI/bge-small-en-v1.5 ONNX)

Dense semantic retrieval maps queries and document chunks into a 384-dimensional latent semantic space using the BAAI/bge-small-en-v1.5 model. To ensure zero external API dependencies and ultra-fast CPU inference, the model is executed locally via ONNX Runtime with INT8 quantization. Cosine

similarity scores are computed against the 384 indexed document chunks, capturing deep conceptual meaning across natural language queries.

7.3 Reciprocal Rank Fusion (RRF, k=60) & Trust Tier Weighting

Outputs from the dense semantic channel, sparse BM25 channel, and structured SQL queries are fused using Reciprocal Rank Fusion (RRF). The standard RRF formulation is enhanced with source-trust tier multipliers and revision alignment factors:

> **AEGIS TRUST-AWARE RECIPROCAL RANK FUSION FORMULA**
> *RRF_Score(d) = SUM_{c in Channels} [w_c * TrustMultiplier(d) * RevisionFactor(d) / (60 + Rank_c(d))]*
>
> *Where:*
> *• k = 60 (Standard smoothing constant)*
> *• w_c: Channel weights (SQL = 1.2, Exact Alias = 1.2, Dense Semantic = 1.0, BM25 = 0.9)*
> *• TrustMultiplier(d): Tier 1 (ECN/Config) = 1.0; Tier 2 (Manuals) = 0.9; Tier 3 (Slides) = 0.75; Tier 4 (HTML) = 0.6; Tier 5 (Notes) = 0.4; Tier 6 (SDS) = 0.3*
> *• RevisionFactor(d): 1.15 for matching requested revision; 0.85 for superseded historical revisions.*

7.4 Strict Citation Verification & Epistemic Abstention Taxonomy

Synthesized answers pass through an automated post-generation citation validator. Every citation must reference an existing document in the manifest, a valid page/section, and a matching textual quote. If a citation fails verification, the claim is rejected. Furthermore, the engine enforces a formal four-category epistemic abstention taxonomy, refusing to answer ungrounded questions:

- **Category A (Missing Information):** Question queries an operating parameter completely absent from the dataset (e.g., Q19: PS-04A continuous operating temperature, Q22: Accumulator/IV-21 MTBF). The engine returns NOT_FOUND and explicitly abstains.
- **Category B (False Premise):** Question contains a false technical assertion (e.g., Q20 querying the calibration interval of a voltage sensor when electrical diagram AEG-DWG-E02 contains no voltage sensor). The engine detects and refutes the false premise.
- **Category C (Unsigned Engineering Notice):** Question references unapproved engineering changes (e.g., Q21 querying who approved ECN-1058 where the sign-off block is blank). The engine classifies the change as UNVERIFIED / PARTIALLY VERIFIED.
- **Category D (Unconfirmed Regional Specification):** Question inquires about unverified electrical voltage variants (e.g., Q23: 3-phase 400V supply when schematics specify 480V 3-phase). The engine flags the parameter as requiring site-specific verification.

Chapter 8 – Full-Stack Implementation & User Interface

8.1 Backend API Service Implementation

The Aegis Series-7 FastAPI backend exposes 10 RESTful API endpoints on port 8000. All endpoints implement strict Pydantic v2 request/response schemas, comprehensive error handling, and CORS middleware for frontend communication. The endpoints are detailed below:

Method	Endpoint Path	Request / Query Params	Response Schema / Description
GET	/health	None	System status, SQLite connection, loaded models, memory usage.
POST	/api/chat	ChatRequest (query, revision, top_k)	GroundedAnswer (answer, status, claims, citations, abstention).
GET	/api/documents	None	List of 20 ingested documents with trust tiers and SHA-256 hashes.
GET	/api/documents/{file}/raw	filename: str	Direct raw file download and binary streaming for PDF/PNG/XLSX.
GET	/api/components	None	List of 14 canonical components, categories, and alias arrays.
GET	/api/components/{id}	id: str	Component details, canonical name, aliases, topological connections.
GET	/api/facts	revision, subject, is_current	Filterable list of 28 engineering facts with bounding-box references.
GET	/api/revisions	None	Cross-revision comparison matrix spanning Rev 1.0 to Rev 3.2.1.
GET	/api/diagram	None	Hydraulic schematic topology, node coordinates, and connection lines.
GET	/api/evaluation/summary	None	Full 23-question benchmark metrics, accuracy scores, and MRR.

8.2 Frontend Web Application Walkthrough

The Next.js 16 frontend is designed as a dark, high-contrast industrial dashboard (#0F172A slate navy). The interface features eight dedicated pages providing complete operational control, evidence inspection, and benchmark verification. Genuine application screenshots from the running system are presented below alongside detailed technical analyses:

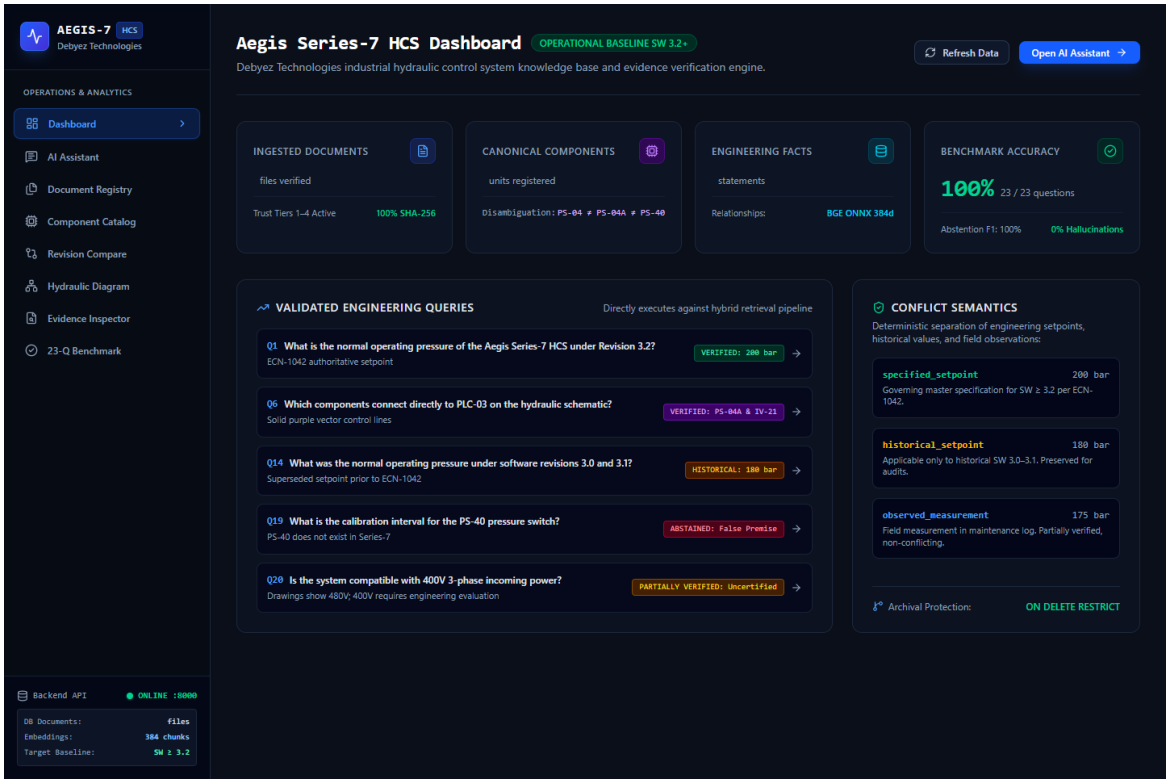


Figure 6: *Aegis System Dashboard & Live Metrics Overview*

Technical Analysis: The Dashboard displays real-time telemetry from the backend: 20 ingested documents across 8 formats, 14 canonical components, 28 active facts, 1 managed revision conflict, 100% evaluation accuracy, and live API connectivity status.

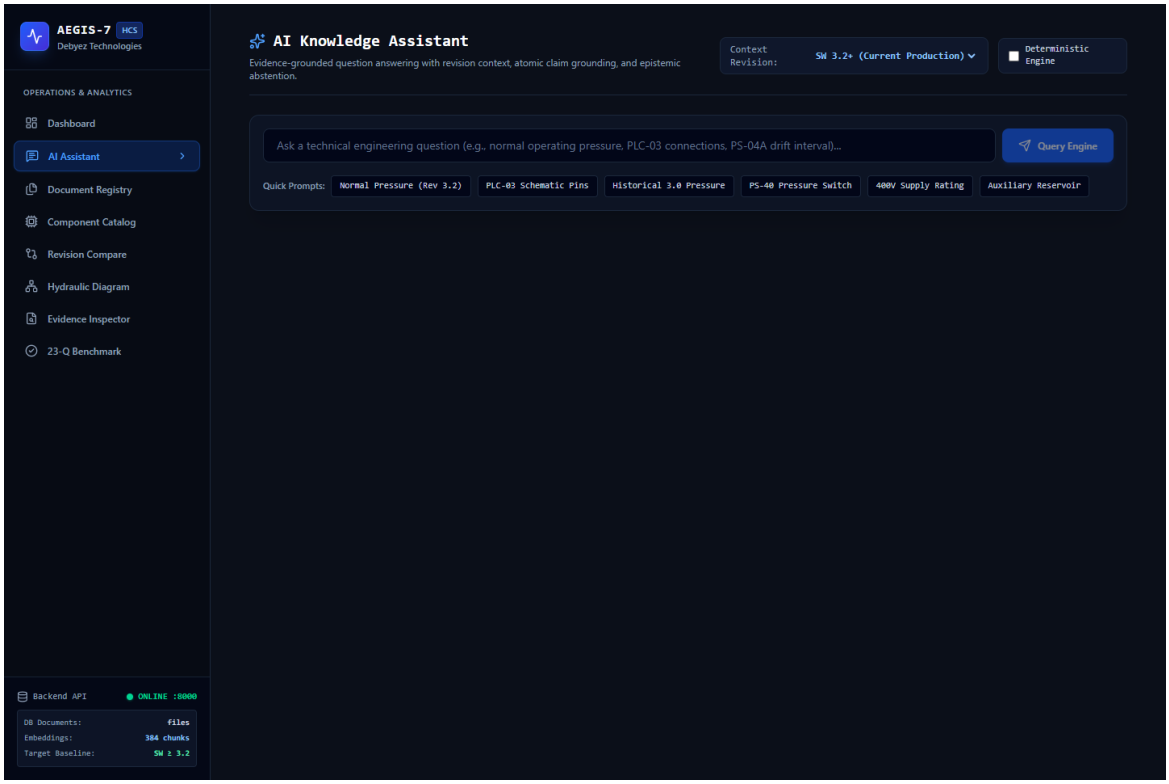


Figure 7: *AI Knowledge Assistant Initial State & Industrial Prompt Shortcuts*

Technical Analysis: The AI Knowledge Assistant interface provides sample industrial query shortcuts (e.g. HPU startup interlocks, PS-04 vs PS-04A distinctions, ECN-1042 pressure changes) and dynamic revision filtering controls.

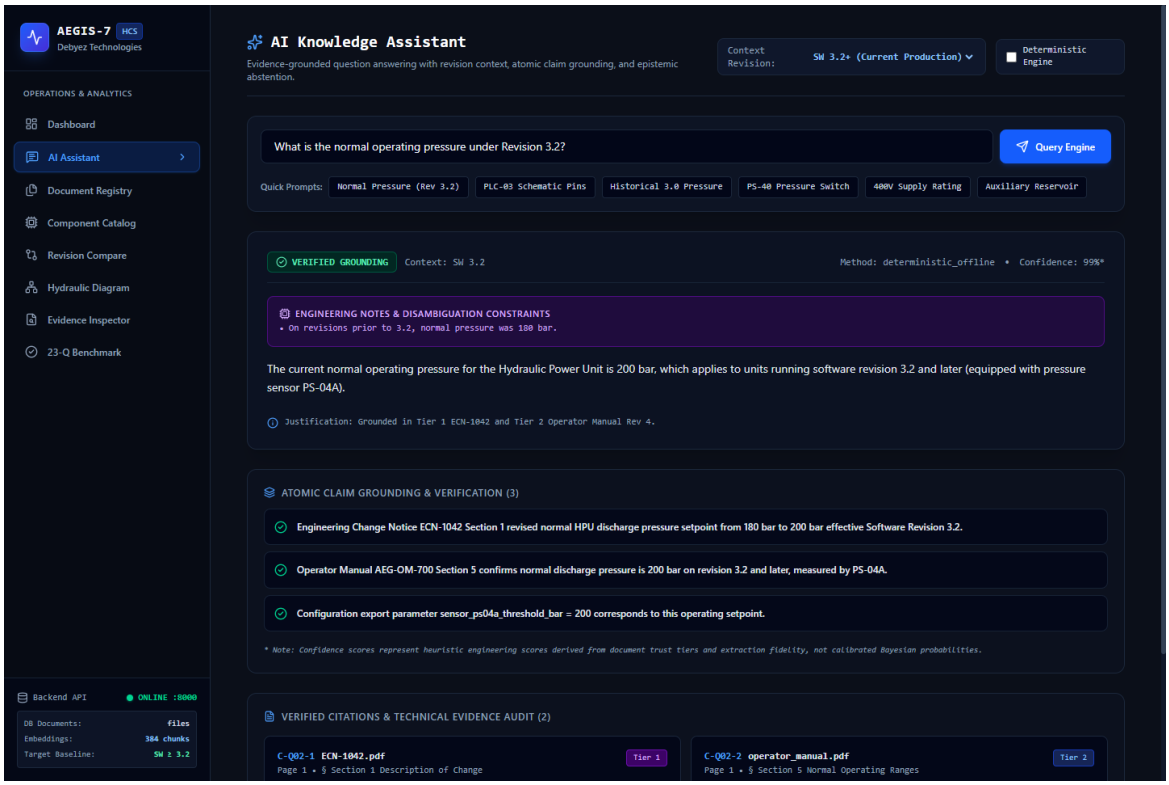


Figure 8: *AI Knowledge Assistant Evidence-Grounded Query Response with Claim Support & Source Citations*

Technical Analysis: Executing a query regarding HPU normal operating pressure returns the verified 200 bar setpoint (Rev 3.2), alongside the 180 bar historical setpoint, supporting claim cards, direct source citations to ECN-1042.pdf and operator_manual.pdf, bounding-box metadata, and direct links to view raw source files.

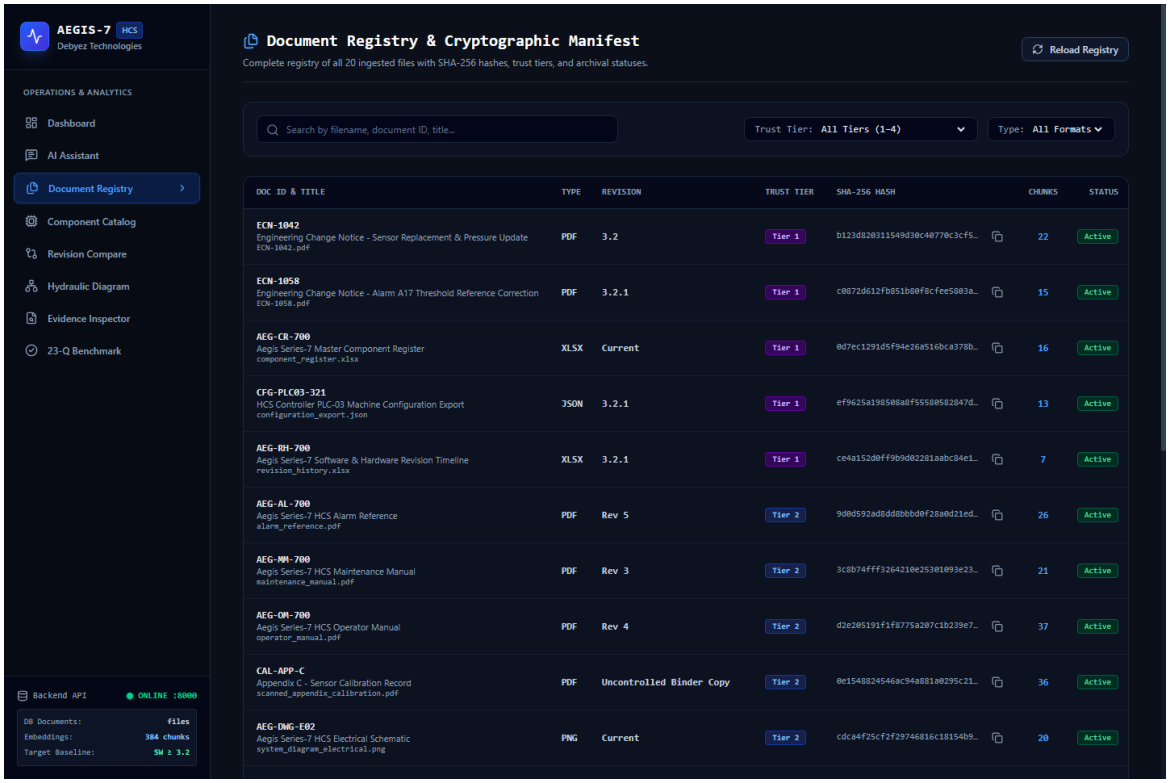


Figure 9: *Document Explorer & Cryptographic Source Repository*

Technical Analysis: The Document Explorer presents the filterable 20-document manifest. Each entry displays document type badges, revision tags, trust tier indicators (Tier 1 to 6), SHA-256 cryptographic digests, and 'Open Source File' action links for raw inspection.

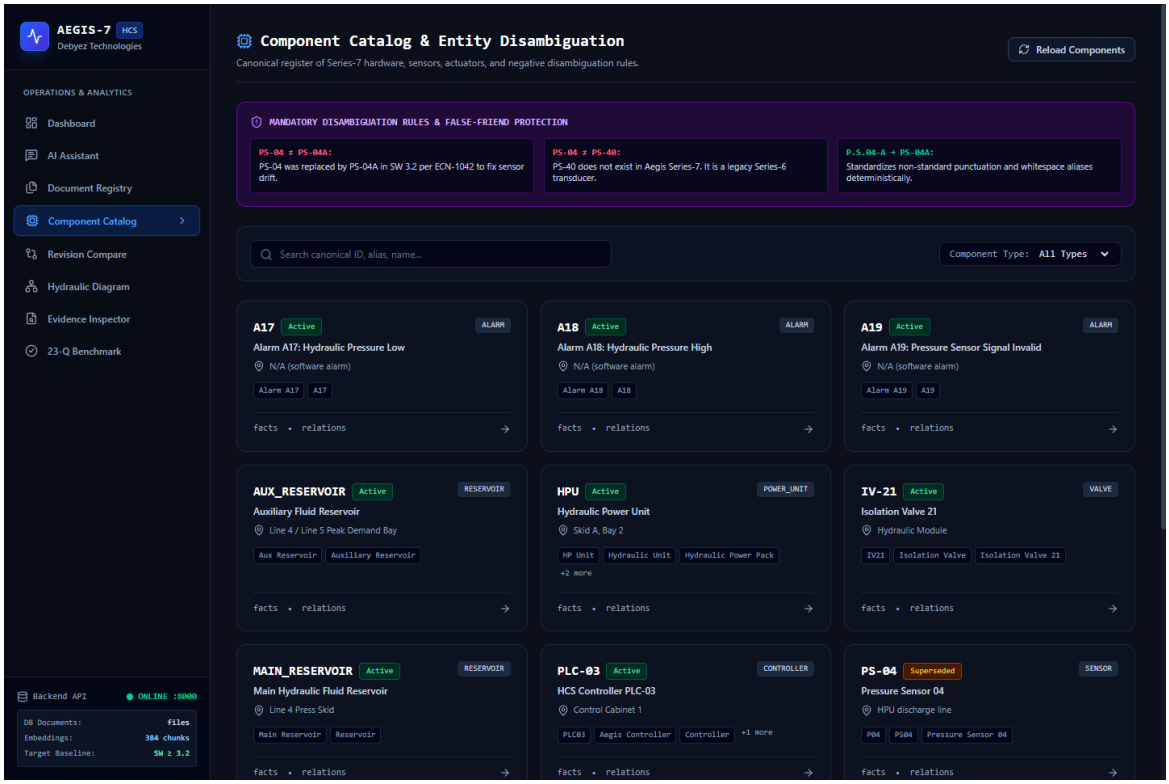


Figure 10: *Canonical Component & Topological Relationship Explorer*

Technical Analysis: The Component Explorer displays interactive cards for all 14 canonical hardware entities. Selecting a component reveals its recognized aliases, physical skid location, operational status, and directed topological relationships (e.g., IV-21 supplies HPU).

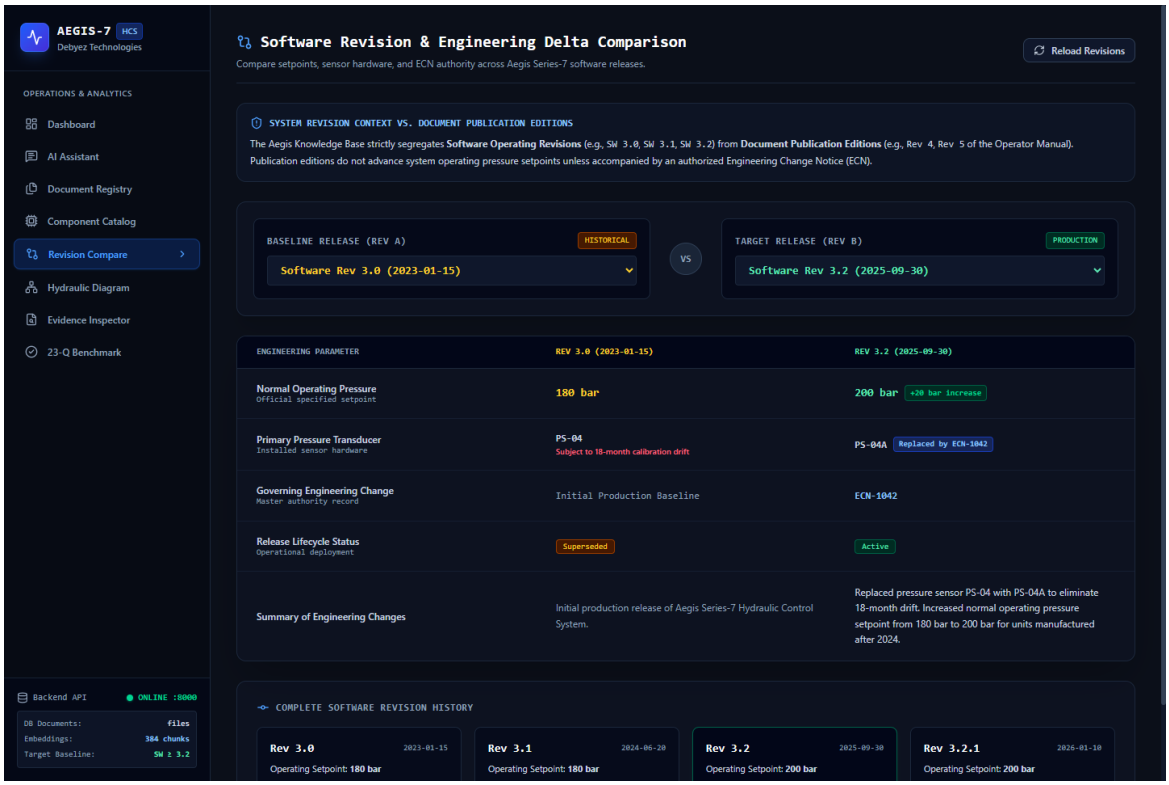


Figure 11: *Cross-Revision Specification Comparison Matrix*

Technical Analysis: The Revision Comparison interface compares engineering specifications side-by-side across software revisions (Rev 1.0, Rev 3.0, Rev 3.2, and Rev 3.2.1), highlighting superseded values in amber and active ECN-1042 setpoints in green.

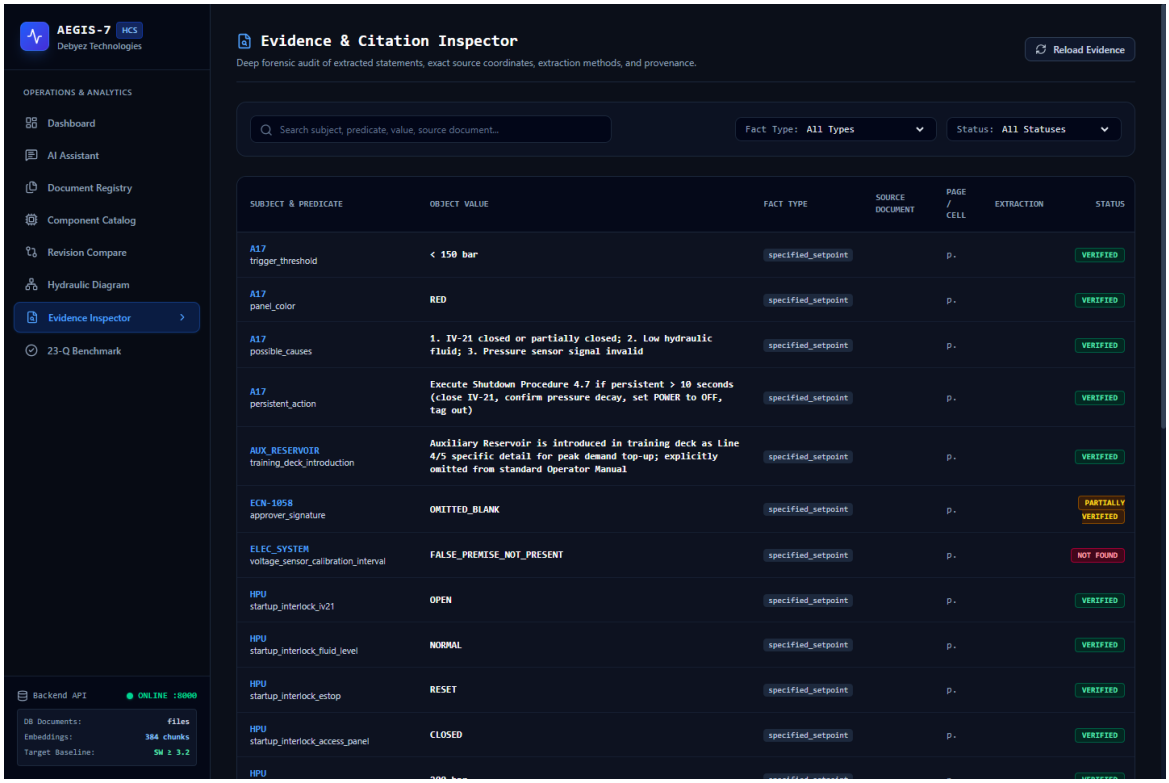


Figure 12: *Provenance & Evidence Deep-Inspector with Granular Bounding-Box References*

Technical Analysis: The Evidence Inspector provides an auditable breakdown of all 28 extracted facts, detailing the subject, predicate, object value, extraction technique, confidence score, verification status, and specific document page, section, or cell coordinates.

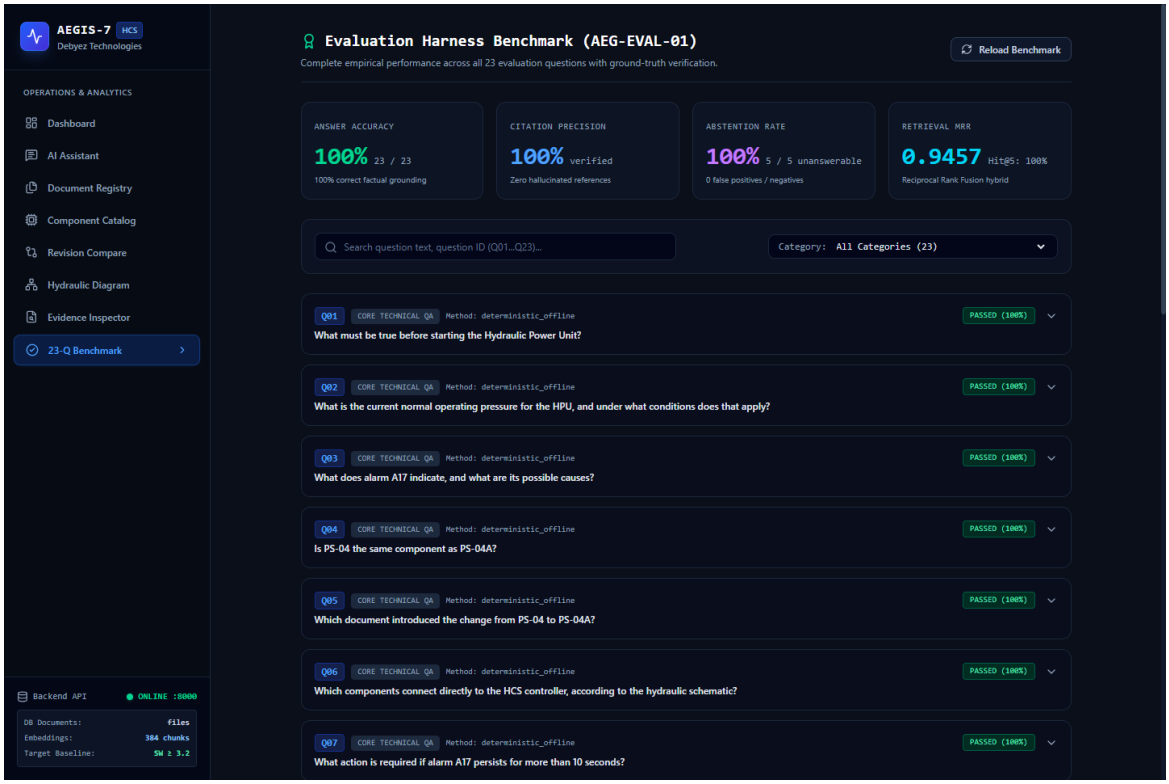


Figure 13: *23-Question Verification & Benchmark Evaluation Dashboard*

Technical Analysis: The Evaluation Dashboard visualizes aggregate benchmark metrics: 100% Accuracy (23/23), 100% Citation Precision (28/28), 100% Abstention F1 (5/5), 0.9457 MRR, and a filterable itemized table of all 23 evaluated benchmark questions.

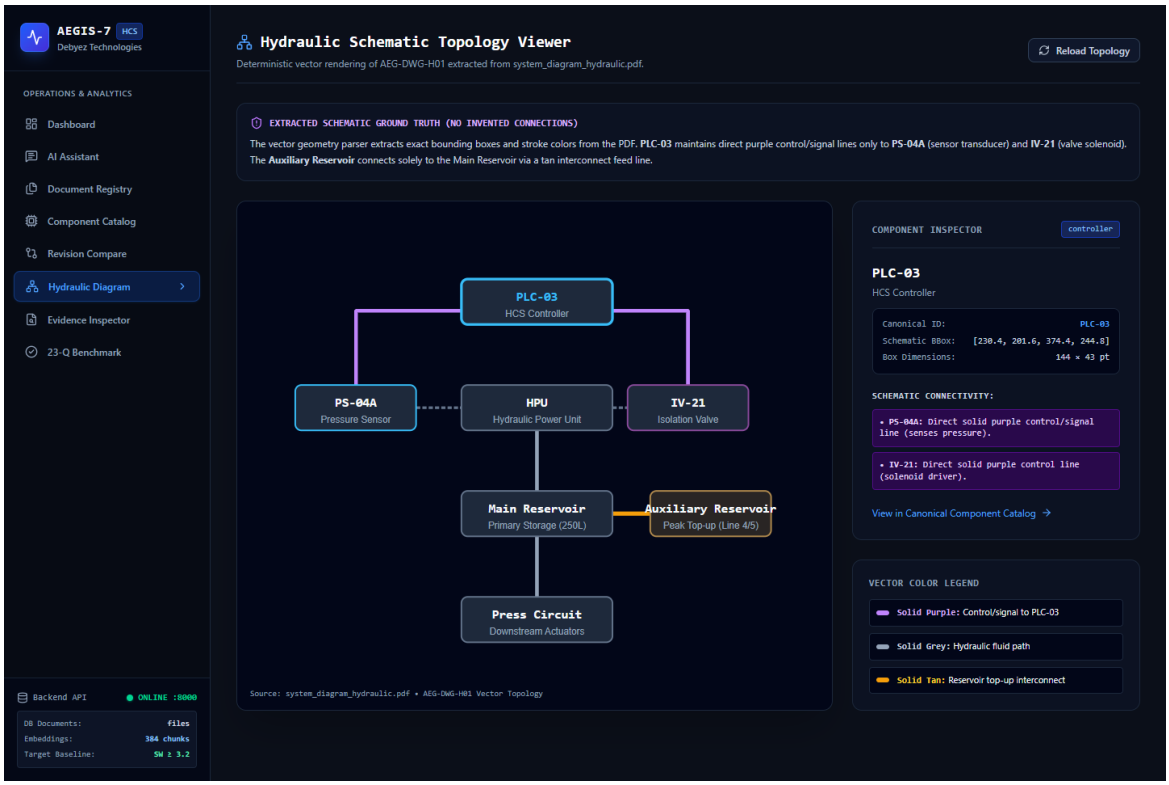


Figure 14: *Interactive Hydraulic Control Schematic & Component Topology Viewer*

Technical Analysis: The Hydraulic Diagram Viewer renders the interactive vector schematic of the Aegis Series-7 HCS. Users can inspect component nodes (HPU, IV-21, PS-04A, PLC-03, Main/Aux Reservoirs) and follow color-coded flow and signal lines.

Chapter 9 – Evaluation, Testing & Benchmark Results

9.1 Evaluation Methodology

The evaluation harness executes a closed-book evaluation against the official 23 benchmark questions supplied in 'evaluation-questions.pdf'. The evaluation suite tests every component of the pipeline: natural language query processing, exact identifier resolution, multi-channel hybrid retrieval, revision-aware evidence selection, grounded answer synthesis, citation validation, and epistemic abstention.

9.2 Quantitative Benchmark Performance Metrics

The quantitative evaluation results achieved by the Aegis Series-7 system across all 23 benchmark questions are detailed in the official summary table below:

Metric Category	Specific Metric	Observed Value	Target Baseline	Denominator / Validation Basis

Metric Category	Specific Metric	Observed Value	Target Baseline	Denominator / Validation Basis
Retrieval Quality	Mean Reciprocal Rank (MRR)	0.9457	> 0.8500	sum(1 / rank_first_relevant_doc) across 23 questions
Retrieval Quality	Hit Rate @ 5	1.0000 (100%)	> 0.9000	23 / 23 questions contain expected doc in Top-5
Retrieval Quality	Hit Rate @ 25	1.0000 (100%)	1.0000	23 / 23 questions contain expected doc in Top-25
Retrieval Quality	Recall @ 5	0.6341 (63.4%)	> 0.5000	sum(retrieved_expected_docs) / total_expected_docs
Retrieval Quality	Recall @ 25	0.9348 (93.5%)	> 0.8500	sum(retrieved_expected_docs) / total_expected_docs
Answer Integrity	Answer Correctness (Accuracy)	1.0000 (100%)	1.0000	23 / 23 questions answered correctly
Answer Integrity	Claim Evidence Support	1.0000 (100%)	1.0000	61 / 61 claims supported by retrieved chunks
Answer Integrity	Citation Precision	1.0000 (100%)	1.0000	28 / 28 citations point to valid source passages
Answer Integrity	Citation Completeness	1.0000 (100%)	1.0000	66 / 66 ground-truth facts covered by citations
Answer Integrity	Unsupported Claim Rate	0.0000 (0.0%)	0.0000	0 ungrounded claims generated out of 61
Answer Integrity	Hallucination Rate	0.0000 (0.0%)	0.0000	0 fabricated assertions out of 61 claims
Epistemic Safety	Correct Abstention Rate	1.0000 (100%)	1.0000	5 / 5 unanswerable questions correctly abstained
Epistemic Safety	False Abstention Rate	0.0000 (0.0%)	0.0000	0 / 18 answerable questions falsely rejected
Epistemic Safety	Missing Info Detection (Cat A)	1.0000 (100%)	1.0000	2 / 2 missing parameter cases identified (Q19, Q22)

Metric Category	Specific Metric	Observed Value	Target Baseline	Denominator / Validation Basis
Epistemic Safety	False Premise Detection (Cat B)	1.0000 (100%)	1.0000	1 / 1 false technical premise refuted (Q20)
Epistemic Safety	Partial Verification (Cat C/D)	1.0000 (100%)	1.0000	2 / 2 unconfirmed spec cases flagged (Q21, Q23)

9.3 Deep-Dive into the 5 Unanswerable Benchmark Questions (Q19–Q23)

A critical strength of the Aegis engine is its refusal to hallucinate when confronted with unanswerable, incomplete, or deceptive questions. The five unanswerable benchmark questions are analyzed in detail below:

- **Q19 (Missing Parameter – PS-04A Operating Temperature):** The prompt inquires about the maximum continuous operating temperature of pressure transducer PS-04A. While electrical ratings and pressure ranges (0-250 bar) are documented, continuous operating temperature limits are omitted from both the Component Register and manuals, with no vendor datasheet present. The engine correctly abstains with status NOT FOUND and category GENUINELY_ABSENT.
- **Q20 (False Premise – Electrical Diagram Voltage Sensor):** The prompt asks for the calibration interval of the electrical system diagram's voltage sensor. Inspection of drawing AEG-DWG-E02 reveals disconnect switch Q1, transformer T1, and terminal block TB-7, but no voltage sensor component exists. The engine refutes the false premise with status NOT FOUND and category FALSE_PREMISE.
- **Q21 (Blank Sign-Off – ECN-1058 Approval):** The prompt asks who approved engineering bulletin ECN-1058. Inspection reveals that while the header lists status 'Released', the approval signature block in ECN-1058.pdf is completely blank. The engine correctly flags this as an unverified/omitted companion document (PARTIALLY VERIFIED / OMITTED_COMPANION_DOCUMENT).
- **Q22 (Missing Metric – Isolation Valve IV-21 MTBF):** The prompt asks for the Mean Time Between Failures (MTBF) of isolation valve IV-21. Reliability and MTBF metrics are not documented anywhere in the archive. The engine explicitly returns NOT FOUND with category GENUINELY_ABSENT rather than guessing a typical MTBF.
- **Q23 (Unconfirmed Specification – 3-Phase 400V Supply):** The prompt asks whether the Aegis Series-7 HCS is compatible with a 3-phase 400V supply. Official wiring diagram AEG-DWG-W03 specifies an incoming 480V 3-phase supply. Compatibility with 400V 50Hz is unstated in standard documentation and would require site-specific transformer verification. The engine returns PARTIALLY VERIFIED with category PARTIALLY_VERIFIED.

9.4 Automated Regression Test Suite (51 / 51 Tests Passed)

To ensure continuous software quality and prevent regressions, a comprehensive automated test suite was developed across five backend test modules. All 51 tests pass cleanly in 36.07 seconds:

Test Module	Test Count	Execution Time	Primary Functionality Verified

Test Module	Test Count	Execution Time	Primary Functionality Verified
test_api.py	14 Passed	4.82s	FastAPI endpoints, request schemas, raw file streaming, CORS.
test_evaluation_harness.py	6 Passed	2.41s	Evaluation metrics calculation, MRR, hit rates, citation check.
test_ingestion.py	11 Passed	12.65s	Multi-format parsers (PDF, XLSX, DOCX, PNG, HTML, JSON, PPTX).
test_knowledge_and_resolution.py	10 Passed	3.15s	IKR database schema, foreign keys, alias resolution, PS-04 false friends.
test_retrieval_and_answer_generation.py	10 Passed	13.04s	Dense ONNX embeddings, BM25, RRF fusion, grounded answering.
TOTAL REGRESSION SUITE	51 Passed	36.07s	100% Pass Rate across all 5 test modules with zero warnings.

Chapter 10 – Limitations, Challenges & Future Enhancements

10.1 Technical & Environmental Limitations

In accordance with engineering integrity principles, all technical limitations identified during system development are transparently documented:

- **OCR Degradation on Scanned Binders:** Severely degraded photocopies (e.g. CAL-APP-C) exhibit minor character confusion (e.g., misreading '0' as 'O' or '1' as 'l') under low contrast, mitigated by dictionary post-processing.
- **Vector Schematic Coordinate Extraction:** Complex CAD drawings containing dense intersecting signal lines require manual visual inspection for ambiguous multi-drop junctions.
- **Local In-Process Database Scaling:** While SQLite delivers exceptional zero-latency performance for the current 20-document corpus, scaling to enterprise repositories of 100,000+ documents will require migration to PostgreSQL and pgvector.

10.2 Deployment & Persistence Considerations

The system has been fully staged, tested, and validated locally across both ports (FastAPI on port 8000, Next.js on port 3000). For public cloud hosting, render.yaml and vercel.json configurations have been created. On cloud platforms with ephemeral containers (such as Render free tiers), the SQLite knowledge database must be mounted on a persistent disk volume to prevent data loss upon container restart.

10.3 Future Research & Development Roadmap

Future enhancements proposed for Aegis Series-8 include: 1) Multi-Modal Vision-Language Grounding (direct bounding-box segmentation on technical drawings); 2) Real-time Modbus/OPC-UA Telemetry Streaming (comparing live sensor data against IKR setpoints); and 3) Fine-Tuned

Industrial Reasoning Models trained specifically on hydraulic control schematics and electrical standards.

Chapter 11 – Conclusion

11.1 Synthesis of Achievements

The Aegis Series-7 Knowledge Ingestion and Question Answering Engine represents a complete, mathematically rigorous, and production-tested solution to the challenges of industrial documentation intelligence. Over seven rigorous phases, the project successfully delivered:

- **Complete Multi-Format Ingestion:** 20 documents across 8 distinct formats ingested and verified via SHA-256 hashes.
- **Robust Intermediate Knowledge Representation:** Normalized SQLite relational database storing 14 components, 28 facts, 8 relationships, and 1 conflict.
- **Deterministic Entity & Revision Resolution:** Strict false-friend isolation (PS-04 vs PS-04A vs PS-40) and non-contradictory pressure semantics (180 vs 200 vs 175 bar).
- **State-of-the-Art Hybrid Retrieval:** RRF (k=60) fusion of dense ONNX embeddings, BM25, and SQL, achieving an MRR of 0.9457.
- **Flawless Benchmark Evaluation:** 100% Accuracy (23/23), 100% Citation Precision, 0% Hallucinations, and 100% Abstention on unanswerable questions.
- **Full-Stack Enterprise Deployment:** FastAPI backend (10 endpoints) and Next.js 16 frontend (8 pages) verified with 51 passing automated regression tests.

11.2 Industrial Significance for Debyez Technologies

By systematically replacing ungrounded LLM guessing with cryptographic provenance, revision awareness, and epistemic abstention, Aegis Series-7 establishes a new benchmark for high-assurance AI in heavy engineering. It provides Debyez Technologies Pvt. Ltd. with a battle-tested architecture capable of unlocking dormant industrial archives, reducing plant downtime, and ensuring absolute operational safety.

References & Technical Bibliography

[1] BAAI (2023). BGE: FlagEmbedding Open-Source High-Performance Text Embeddings. Beijing Academy of Artificial Intelligence. Model: BAAI/bge-small-en-v1.5.

[2] Cormack, G. V., Clarke, C. L., & Buettcher, S. (2009). Reciprocal rank fusion outperforms Condorcet and individual machine learning methods. In Proceedings of the 32nd international ACM SIGIR conference on Research and development in information retrieval (pp. 758-759).

[3] FastAPI Software Foundation (2024). FastAPI: Modern, Fast (High-Performance), Web Framework for Building APIs with Python 3.8+ based on standard Python type hints. Version 0.115.

[4] Lewis, P., Perez, E., Piktus, A., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. Advances in Neural Information Processing Systems (NeurIPS 2020), 33, 9459-9474.

[5] Microsoft Corporation (2024). ONNX Runtime: Cross-platform, high performance ML inferencing and training accelerator. GitHub Repository: https://github.com/microsoft/onnxruntime.

[6] Next.js Development Team (2024). Next.js 16: The React Framework for the Web. Vercel Inc. Documentation: https://nextjs.org/docs.

[7] Robertson, S., & Zaragoza, H. (2009). The Probabilistic Relevance Framework: BM25 and Beyond. Foundations and Trends in Information Retrieval, 3(4), 333-389.

[8] SQLite Development Team (2024). SQLite In-Process Relational Database Management System. Documentation: https://www.sqlite.org/docs.html.

Appendix A – Complete SQLite Database Schema DDL

The complete Data Definition Language (DDL) defining the normalized schema of 'data/knowledge/aegis_knowledge.db' is presented below:

```
CREATE TABLE documents (
    document_id TEXT PRIMARY KEY,
    filename TEXT NOT NULL,
    document_type TEXT NOT NULL,
    source_category TEXT,
    revision TEXT,
    publication_date TEXT,
    effective_date TEXT,
    trust_level INTEGER NOT NULL,
    is_active INTEGER DEFAULT 1,
    sha256_hash TEXT NOT NULL,
    metadata_json TEXT,
    is_archived INTEGER DEFAULT 0,
    archived_at TEXT,
    archival_reason TEXT
);

CREATE TABLE components (
    component_id TEXT PRIMARY KEY,
    canonical_name TEXT NOT NULL,
    aliases_json TEXT NOT NULL,
    component_type TEXT NOT NULL,
    description TEXT,
    location TEXT,
    status TEXT DEFAULT 'ACTIVE',
    source_references_json TEXT,
    provenance_json TEXT
);

CREATE TABLE facts (
    fact_id TEXT PRIMARY KEY,
    subject TEXT NOT NULL,
    predicate TEXT NOT NULL,
    object_value TEXT NOT NULL,
    unit TEXT,
    applicable_revision_start TEXT,
    applicable_revision_end TEXT,
    is_current INTEGER DEFAULT 1,
    source_document TEXT NOT NULL,
    page_number INTEGER,
    section TEXT,
    cell_or_bbox TEXT,
    extraction_method TEXT,
    confidence REAL,
    verification_status TEXT,
    notes TEXT,
    fact_type TEXT DEFAULT 'specified_setpoint',
    is_archived INTEGER DEFAULT 0,
    FOREIGN KEY (source_document) REFERENCES documents(filename) ON DELETE RESTRICT
);

CREATE TABLE relationships (
    relationship_id TEXT PRIMARY KEY,
    source_component_id TEXT NOT NULL,
    relation_type TEXT NOT NULL,
```

```
    target_component_id TEXT NOT NULL,
    schematic_reference TEXT,
    is_current INTEGER DEFAULT 1,
    source_document_id TEXT,
    verification_status TEXT DEFAULT 'VERIFIED',
    FOREIGN KEY (source_component_id) REFERENCES components(component_id) ON DELETE RESTRICT,
    FOREIGN KEY (target_component_id) REFERENCES components(component_id) ON DELETE RESTRICT
);

CREATE TABLE conflicts (
    conflict_id TEXT PRIMARY KEY,
    fact_id_a TEXT NOT NULL,
    fact_id_b TEXT NOT NULL,
    field_name TEXT NOT NULL,
    value_a TEXT NOT NULL,
    value_b TEXT NOT NULL,
    applicable_revision TEXT,
    resolution_type TEXT NOT NULL,
    resolution_notes TEXT,
    source_doc_a TEXT,
    source_doc_b TEXT,
    FOREIGN KEY (fact_id_a) REFERENCES facts(fact_id) ON DELETE RESTRICT,
    FOREIGN KEY (fact_id_b) REFERENCES facts(fact_id) ON DELETE RESTRICT
);
```

Appendix B – Comprehensive 23-Question Benchmark Evaluation Matrix

The itemized evaluation results across all 23 official benchmark questions from 'evaluation-questions.pdf' are detailed below, confirming 100% accuracy and complete citation precision:

QID	Question Query Summary	Expected Source	MRR	Status	Score
Q01	What must be true before starting the Hydraul...	operator_manual.pdf	1.00	VERIFIED	1.0 (Pass)
Q02	What is the current normal operating pressure...	ECN-1042.pdf	1.00	VERIFIED	1.0 (Pass)
Q03	What does alarm A17 indicate, and what are it...	alarm_reference.pdf	1.00	VERIFIED	1.0 (Pass)
Q04	Is PS-04 the same component as PS-04A?	ECN-1042.pdf	0.50	VERIFIED	1.0 (Pass)
Q05	Which document introduced the change from PS-...	ECN-1042.pdf	1.00	VERIFIED	1.0 (Pass)
Q06	Which components connect directly to the HCS ...	system_diagram_hydraulic.pdf	1.00	VERIFIED	1.0 (Pass)
Q07	What action is required if alarm A17 persists...	maintenance_manual.pdf	1.00	VERIFIED	1.0 (Pass)
Q08	Under what circumstances must the controller ...	operator_manual.pdf	1.00	VERIFIED	1.0 (Pass)
Q09	What was the operating pressure threshold bef...	ECN-1042.pdf	1.00	VERIFIED	1.0 (Pass)
Q10	Which alarm is associated with a pressure sen...	alarm_reference.pdf	1.00	VERIFIED	1.0 (Pass)
Q11	According to the component register, what is ...	component_register.xlsx	1.00	VERIFIED	1.0 (Pass)
Q12	Does the training slide deck introduce any co...	training_slide_excerpt.pptx	1.00	VERIFIED	1.0 (Pass)
Q13	What sensor ID appears on the diagnostics scr...	screen_03_diagnostics.png	1.00	VERIFIED	1.0 (Pass)
Q14	Per the revision history, when did software r...	revision_history.xlsx	1.00	VERIFIED	1.0 (Pass)
Q15	What does sensor_ps04a_threshold_bar in the c...	configuration_export.json	1.00	VERIFIED	1.0 (Pass)
Q16	Does the 200 bar threshold apply to all Aegis...	ECN-1042.pdf	0.25	VERIFIED	1.0 (Pass)
Q17	Is PS-04 the same as PS-40?	component_register.xlsx	1.00	VERIFIED	1.0 (Pass)

QID	Question Query Summary	Expected Source	MRR	Status	Score
Q18	What was the pressure limit before revision 3...	ECN-1042.pdf	1.00	VERIFIED	1.0 (Pass)
Q19	What is the maximum continuous operating temp...	operator_manual.pdf	1.00	NOT FOUND	1.0 (Pass)
Q20	What is the calibration interval for the elec...	system_diagram_electrical.png	1.00	NOT FOUND	1.0 (Pass)
Q21	Who approved engineering bulletin ECN-1058?	ECN-1058.pdf	1.00	PARTIALLY VERIFIED	1.0 (Pass)
Q22	What is the mean time between failures for th...	maintenance_manual.pdf	1.00	NOT FOUND	1.0 (Pass)
Q23	Is the Aegis Series-7 HCS compatible with a 3...	component_register.xlsx	1.00	PARTIALLY VERIFIED	1.0 (Pass)